# Linyi (Tiger) Hou

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## Education

**Ph.D. in Aerospace Engineering** May 2025
University of Illinois Urbana-Champaign
Thesis: Autonomous Lost in Space and Time State Determination using Periodic Variable Stars

**B.S. in Aerospace Engineering, Minor in Computer Engineering** May 2021
University of Illinois Urbana-Champaign

## Skills

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|---|---|
| Technical | celestial navigation, orbit determination, batch filtering, Kalman filtering, Monte Carlo simulation, spacecraft navigation modeling |
| Collaboration | technical/proposal writing, MS Office suite, LaTeX, Git, GitHub, Trello |
| Programming | proficient: Python, C/C++, MATLAB \| beginner: Simulink, Javascript, Julia |

## Experience

**Graduate Researcher, University of Illinois Urbana-Champaign** Jun 2021 – May 2025

- Developed an autonomous optical navigation concept using variable star light curves which delivers timing accuracy within 1 second without needing prior position/time information.
- Collaborated with colleagues to create navigation simulation tools in Python to assess the feasibility of replacing DSN tracking with celestial navigation on OSIRIS-APEX.
- Designed sensor scheduling algorithms for celestial navigation which reduced navigation uncertainty by over 10% compared to existing strategies in Monte Carlo simulation.
- Created a photon time of arrival simulation tool in Python and validated simulation results against the pulsar timing package PINT to within 50 μs.
- Developed a novel initial orbit determination technique using batched sequential range-rate measurements that achieved over 10x better accuracy than state-of-the-art methods.
- Developed a star identification technique using C++ which mitigates stellar aberration effects at relativistic speeds, improving star identification success rate from 38% to 91%.

**Teaching Assistant, University of Illinois Urbana-Champaign** Jun 2021 – May 2024

- Worked with 2-5 faculty and staff to teach lessons, hold office hours, and improve curriculum for courses in flight mechanics, systems engineering, and UAV controls.
- Helped 20+ students formulate and implement UAV state-space model controllers with various sensors, e.g., laser rangefinders, optical flow cameras, and time-of-flight anchors.
- Assisted in developing LQR control-related code and instructional materials for a UAV course using C and Python.

**Software Engineering Intern, Collins Aerospace** May 2019 – Aug 2019

- Migrated an avionics data interpreter from 16-bit to 32-bit using C++ and Python.
- Automated weekly builds and integration in TortoiseSVN using Python.

# Projects

**Collaborative Projects in Flight Dynamics** 2022, 2023

- Planetary entry: Created a 3-DOF entry simulation to assess the flight performance of several novel Apollo-derived entry guidance algorithms at Mars.
- Trajectory optimization: Applied optimal control theory to identify time- and fuel-optimal low-thrust Earth-Mars transfer trajectories.

**Team Lead/Advisor, NASA's RASC-AL Competition** 2019, 2022 – 2023

- Led a team of 10+ students to develop a reusable crewed lunar lander concept (structure, power, thermal, life support, orbital mechanics, budget, etc.).
- Advised two student teams who developed technical requirements and performed trade studies for lunar and Martian in-situ resource utilization architectures.

# Honors and Awards

*Aerospace Engineering Alumni Advisory Board Fellowship*, UIUC 2024
*Best Student Paper*, 46$^{th}$ AAS GN&C Conference, American Astronautical Society 2024
*Yee Memorial Fellowship*, UIUC 2021, 2024
*David and Catherine Thompson Space Technology Scholarship*, AIAA 2020